

EXECUTIVE BRIEFING

Cut your agentic-coding token bill

Coding agents are becoming a top-line AI cost. The lever is choosing the right **model and harness** from evidence -- not defaulting to the most expensive option.

WHAT YOU'LL GET

Three outcomes

A buying guide -- a cost/quality frontier that picks the model by evidence, not reputation.

Measured on your work -- 16 models over 21 real tasks from your own repositories, scored by an independent judge.

Managed spend -- a skill your developers install that picks the model per task, so the saving does not depend on anyone remembering.

THE PROBLEM

Organizations are heading toward trillions of tokens a month -- and coding agents drive a large, growing share of it.

Two choices decide most of that bill -- **which agent (harness)** drives the work and **which model** it uses -- yet they get made on gut feel, or on public leaderboards that may not reflect your real work.

WHY GENERIC NUMBERS MISLEAD

Real coding work doesn't look like a benchmark prompt

50-300

LLM turns per task -- long-horizon, not one-shot

150:1+

input-to-output ratio -- read-heavy, up to 660:1

8x

cost gap between the top model and the cheapest scoring near it

Cost depends on **how** a model drives the task. You need numbers measured on **work that looks like yours**.

THE APPROACH

Measure **quality and **cost** on real coding tasks -- across agents and hosting options -- then let the **cost/quality frontier** pick the model.**

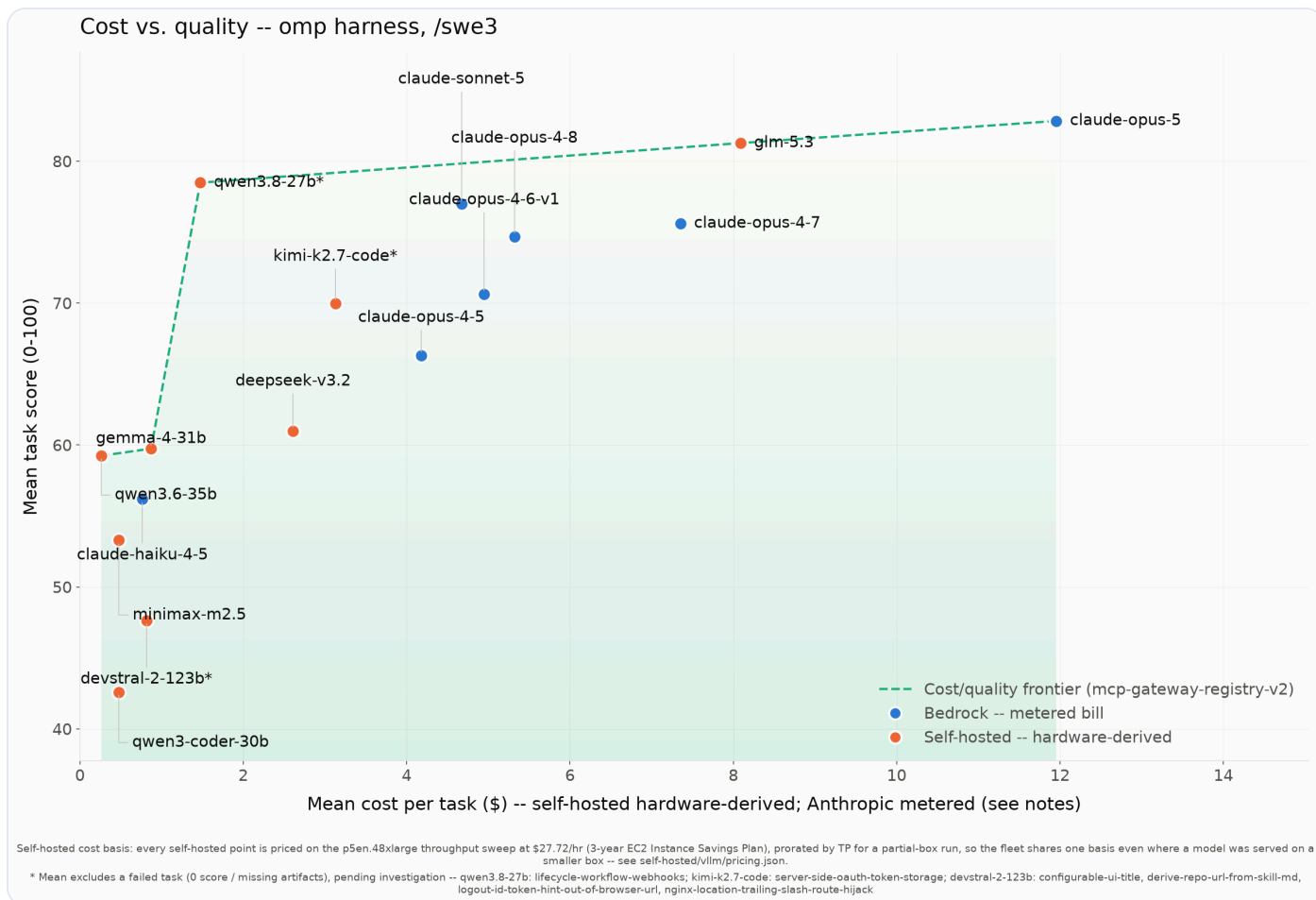
The frontier is the set of models where nothing else is both better and cheaper.
It turns model selection into a defensible, budget-aware decision.

HOW TO READ THE NEXT SLIDE

The **frontier** = the models where nothing else is both better *and* cheaper.

Everything off it is **dominated** -- some frontier model wins on both quality and cost, so you would never pick the other. To choose: pick the quality your task needs, then take the cheapest model on the frontier at that level. **Example:** a 70-point model at \$3.13/task is dominated by a 78-point model at \$1.47/task -- higher score, lower cost -- so it is off the frontier.

Your buying guide: the cost/quality frontier



16 models, 21 real tasks. Higher is better; left is cheaper. The line is the frontier -- the only models worth considering. Everything below it is dominated (something is both better and cheaper).

WHAT TO BUY FOR WHAT

Match the model to the job

Business-critical work -- claude-opus-5, 82.8 at \$11.95 a task. Pay the most, get the most; use it where a wrong result is expensive.

Everyday engineering -- qwen3.8-27b, 78.5 at \$1.47 a task. Within 4.4 points of the best model at an eighth of the cost.

Routine / high-volume edits -- qwen3.6-35b, 59.2 at 26 cents a task, for boilerplate and small fixes you review anyway.

Somebody still has to make that pick, per task. **That is what the next slide takes off them.**

FROM CHART TO DAILY PRACTICE

Measure once, spend less on every task

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Everyone knows the top model is overkill for most tasks. This makes the cheaper choice the automatic one.

--- PLATFORM TEAM a cron job -- runs weekly, unattended ---

1 · MEASURE

self-hosted managed service vendor API

Available models

whatever security approved

Your dataset
tasks from your
repo

Run the benchmark

every model × every task, judged

Your frontier

not a vendor claim, not a public set
that leaked into training data

SHIPS AS THE SWE-ROUTER SKILL

models.json

score + cost, per difficulty tier

allowed-models.txt

the same approved list, now the
filter

route.py

the decision, as code

One `curl` to install. Every
developer reads the same numbers on
the same day.

--- EVERY DEVELOPER every task ---

2 · ROUTE

Claude Code Codex pi -

A task begins

`swe-router` is installed in the
harness -- it engages on its own,
nobody invokes it

How bad if wrong?

→ a floor

How hard?

→ a table

Cheapest model clearing the floor

ranked over what this developer
can actually select

PLATFORM TEAM GETS

Up to 88% less

per task, against running the top model on every task -- measured on your own repo, not claimed.

DEVELOPERS GET

No decision

The right model arrives with the task. Nobody
weighs quality against the bill, twenty times
a day.

A platform team runs the benchmark on a schedule -- a cron job over the models security approved and your own repositories.

The result ships as a skill. Five files install into whichever coding assistant your developers already use.

It engages on its own before a substantial task, names the cheapest model that clears the bar, and stops. The developer switches.

BEDROCK OR SELF-HOSTED

**Works whether you buy tokens on
Amazon Bedrock or self-host open-
weight models on your own GPUs.**

Same tasks, same scoring -- so the trade-off is measured on one footing. At high volume, self-hosting a right-sized open-weight model on committed GPU capacity is a real lever against per-token pricing.

THE TAKEAWAY

Turn coding-agent spend into a managed decision

Stop defaulting to the most expensive model-and-agent for every task.

Choose from the frontier -- measured on your repos, your tasks, your hosting.

Install the skill -- so the cheaper choice happens on every task, without anyone looking anything up.

EXPLORE THE EVIDENCE

See the full benchmark

Open source. Reproduce the frontier on your own code and models.

github.com/aarora79/agentic-coding-harness-benchmarks



Scan for the repository